

22 Ans: C

$$\begin{aligned}\text{From } I &= Q / t \\ &= 6.0 / (2 \times 60) = 0.050 \text{ A}\end{aligned}$$

$$\begin{aligned}\text{From } P &= V I \\ \frac{48}{2 \times 60} &= V (0.050) \\ V &= 8.0 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{Thus, from } E &= IR + Ir \\ 9.0 &= 8.0 + (0.050) r \\ r &= 20 \ \Omega\end{aligned}$$

23 Ans: A